Experiment 7.1: Simulated Radioactive Decay (Method 1)

Notes

Background

This investigation is a simulation of radioactive decay. Radioactive decay is a random event. You will have noticed this by watching or listening to the uneven count rate on a Geiger counter when it is detecting low intensity radiation. If, however, a large number of atoms are disintegrating, the average number of disintegrations per second can be predicted with reasonable accuracy.

The half-life of a radioactive substance is the time taken for its decay rate or activity to decrease by one half. This is also the time taken for the number of undecayed nuclei to decrease by one half.

Aim

To simulate half life using a non-radioactive decay process.

Apparatus (per group)

- 1 packet of M & Ms®
- 1 clean paper cup
- 1 sheet of clean paper
- 1 or more consumers (for discarded M & Ms)

Number of throws	Number of disintegrations	Number of nuclei remaining
0	0	
1		

Pre-Lab

- In this simulation you will use M & Ms to represent radioactive nuclei. A 'decayed' M & M is one which falls with the 'M' facing up. Discuss how you will safely dispose of the wastes from your experiment.
- Set up a table suitable for recording the data from your simulation. An example is shown (*above*).

Lab Notes

- Begin by tipping the M & Ms onto the sheet of paper on the bench-top. Count them and then transfer them to the paper cup. Enter this information in your data table a time t = 0.
- Again tip the M & Ms onto the sheet of paper. This is your first 'throw'.
- Count and remove the 'decayed' nuclei (the M & Ms with the M facing up). Put the remaining 'undecayed' M & Ms back in the paper cup. Enter the data in your table at time t = 1.
- Repeat these steps until all of your sample has decayed.
- Dispose of the 'decayed' M & Ms thoughtfully!
- All groups will pool their data at the end so that you can include a much larger number of starting nuclei (M & Ms) in the final analysis.
- Transfer your data to a table of the whole class's results.

Post-Lab Discussion

1 In this simulation, time is represented by the tipping of the M & Ms onto the paper. Plot a graph of the number of nuclei remaining against 'time-units'.

2 Plot a graph of the class results.

Further Investigation

- How would the results differ if you used a large number of dice instead of M & Ms?